

Ahmadreza Farahani | ML/AI Software Engineer

🌐 ahmadrezafrh.github.io | ✉ amadreza.farahani@outlook.com | 🏠 Turin, Italy (open to relocation)
🌐 linkedin.com/in/amadrezafrh | ☎ +39 351 641 1678 | 📅 4+ Years of Experience

Skills

Agentic AI	LangGraph, LangChain, LangSmith, MCP, agentic RAG (hybrid retrieval, reranking), guardrails, Pydantic
LLM Systems	Transformers, HuggingFace, Ollama, vLLM, TorchServe, Milvus, ChromaDB, pgvector, LoRA, QLoRA
Machine Learning	NumPy, Pandas, scikit-learn, PyTorch, TensorFlow, XGBoost, MLflow, GANs, U-Net, YOLO, ASR
Cloud Technologies	AWS (Bedrock, ECS, EKS, EC2, S3, Lambda, SageMaker, CloudWatch), Docker
Software Development	Python, C/C++, SQL, Bash, Linux, Git, CI/CD, FastAPI, Prometheus, Grafana, Airflow

Professional Experience

AI/ML Software Engineer

Zirak

Turin

11/2024 - Current

- Delivered a multi-hop QA agent on **AWS Bedrock** with **GraphRAG** and Amazon Neptune to query over 1,000 linked documents.
- Launched an ISO document change-analysis product detecting 96% of revision changes; engineered a hybrid BM25 and dense retrieval pipeline into **pgvector** and implemented model routing to cut LLM costs by 65% while maintaining recall.
- Automated 81% of inbound email volume fully on-premise by developing an email observer agent using fine-tuned Llama 3.2 3B (QLoRA) and **LangChain** that achieved a 94% F1 score on structured extraction and routed low-confidence cases to human review.
- Reduced query SQL authoring time for non-technical analysts at **Poste Italiane**'s logistics division with a production NL-to-SQL agent: **LangGraph** routing **MCP** tools across PostgreSQL and MySQL, served on-prem via FastAPI and **Ollama**.
- Created an on-device meeting assistant agent with real-time transcription, diarization, and speaker verification with a **LangGraph** agentic RAG (hybrid retrieval + reranking) plus **LangSmith** regression evals that cut unsupported claims in summaries by 25%.
- Deployed a document-review agent (HuggingFace models served on **vLLM** with **Guardrails**), saving 50 hours/month internally and later licensed to external clients including **Techyon**.
- Developed a custom U-Net architecture to detect invasive algae from multispectral satellite imagery, achieving a +10% F1 improvement over internal XGBoost baseline and cutting manual review by 80% via **Airflow** on AWS SageMaker.
- Built a rail-safety detection and segmentation pipeline for **Blue Engineering** using YOLOv8 and **PyTorch**, deploying the system on **AWS ECS** with Prometheus and Grafana for real-time monitoring.

Visiting Researcher - Master Thesis

Technische Universität Darmstadt | CROSSING

Darmstadt

09/2023 - 11/2024

- Architected a Diffusion **Transformer** for voice conversion (Python, PyTorch) on HPC GPUs, improving speaker similarity by 9% over open-source baselines with fewer sampling steps, enabling real-time conversion | [Blog](#)
- Ran a prototype low-latency inference on AWS ECS, tracking experiments in **MLflow** and monitoring latency with CloudWatch.

ML Software Engineer

PartAI Research Center

Tehran

02/2020 - 03/2022

- Trained an on-device keyword-spotting Android app in Java, using hard negative samples to reduce false positives by 30%.
- Optimized a speaker verification system on **TensorFlow**, fine-tuned on new domains that reduced the Equal Error Rate (EER) by 2%.
- Established Test-Driven Development (TDD) for Persian ASR and NLP services; wrote unit/integration tests reaching 100% coverage and set up **Jenkins** CI/CD pipelines.
- Mentored new team members on clean, production-ready Python (testing, logging, design patterns), improving quality and safety.
- Implemented deep neural networks (from research papers) for speaker identification and speaker verification, packaged with **Docker**.

Internship Experience

Associate Generative AI Software Engineer

AROL S.p.A

Canelli

09/2022 - 04/2023

- Designed a synthetic machinery data generator (**GAN**, TensorFlow, trained on **AWS SageMaker**) packaged with MongoDB and Docker Compose plus REST API docs, so customers run it locally without exposing proprietary data.

Computer Vision Research Intern - Bachelor Thesis

Megamouj Co. | Tehran University of Science and Technology

Tehran

09/2019 - 02/2020

- Automated a traffic-flow monitoring system (C++, Python, **OpenCV**, YOLOv3) for vehicle detection, tracking, and flow estimation, reducing manual video review in a national-scale project | [Blog](#)
- Engineered data pipelines with **NumPy** and Pandas to structure annotation datasets and computed evaluation metrics in **scikit-learn** to validate system reliability.

Education

M.Sc. Computer Engineering - Artificial Intelligence & Data Analytics

Politecnico di Torino

Turin, Italy

2022 - 2025

- Courses:** Advanced Machine Learning, System and Device Programming, Software Engineering, Web Applications
- Thesis:** RobVC: A Robust zero-shot End-to-End Self Supervised Voice Conversion (Funded by CROSSING Co. in Darmstadt) | [Blog](#)
- GPA:** 3.42/4.00

B.Sc. Electrical and Computer Engineering - Telecommunications

Iran University of Science and Technology

Tehran, Iran

2016 - 2021

- Courses:** Programming, Digital Signal Processing, Linear Algebra, Linear Control Systems, Communications
- Thesis:** Deep Learning-Based Vehicle Detection and Traffic Flow Analysis (Funded by Megamouj Co.) | [Blog](#)
- GPA:** 2.96/4.00